

# Sagor Das

BIOMEDICAL ENGINEERING STUDENT & RESEARCHER



## CONTACT

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## ABOUT MYSELF

Biomedical Engineering student at Military Institute of Science and Technology (MIST) with hands-on research experience at the National Institute of Biotechnology (NIB). Skilled in molecular biology, microbial biotechnology, bioinformatics, and computational drug discovery. Experienced in microfluidics, finite element analysis (ANSYS, COMSOL), biomechanical modeling (OpenSim), and AI-based healthcare systems. Actively engaged in interdisciplinary biomedical research combining wet lab and computational approaches. Seeking opportunities in biomedical research, diagnostics, and healthcare innovation.

## Education

- **Bachelor of Science (B.Sc.) in Biomedical Engineering**  
Military Institute of Science and Technology  
**Year:** 2026  
Relevant Courses: Molecular Biology, Bioinformatics, Medical Instrumentation, Biomaterials, Signal Processing, Microfluidics, Biomechanics  
**Result:** CGPA 3.66 out of 4
- **Higher Secondary Certificate (HSC) — Science**  
Subjects included Physics, Chemistry, and Biology  
Bhasha Soinik Ajit Guha College, Comilla, Bangladesh  
**Year:** 2021  
**Result:** GPA 5.00 out of 5
- **Secondary School Certificate (SSC) — Science**  
Subjects included Physics, Chemistry, and Biology  
**Year:** 2019  
**Result:** GPA 5.00 out of 5

## PROFESSIONAL EXPERIENCE

### Research Assistant

- MIST (Military Institute of Science and Technology)  
Duration: May 12, 2025 – June 4, 2025

## RESEARCH INTERESTS

- Biomedical Engineering
- Medical Robotics & Cyber-Physical Systems
- Bio-Materials & Microfluidics
- AI in Healthcare & Digital Diagnostics
- Biosignal Processing & Health Informatics
- Cancer Diagnostics & Computational Biology
- Soft Matter & Bioinspired Robotics
- Point-of-Care Technologies & Wearables

## B.SC. THESIS

### Development of a Microfluidic Device for Detection of Anti-dsDNA Autoantibodies and Therapeutic Screening

**Status:** Complete

- Designed a microfluidic biosensor for detection of anti-dsDNA autoantibodies, a key biomarker for Systemic Lupus Erythematosus (SLE)
- Integrated computational drug screening and binding affinity prediction methods
- Focused on early-stage diagnosis of autoimmune diseases using point-of-care technology

# PROJECTS

## 1. AI-Powered Web-Based Rehabilitation Exercise Tracker (2025–2026)

- Built AI-based rehabilitation tracking system using pose and hand detection
- Used computer vision and MediaPipe for real-time posture correction
- Designed for remote physiotherapy applications
- **Tools:** HTML, CSS, JS, PYTHON

## 2. Digital Heart Murmur Detection System (2024)

- Developed signal processing-based stethoscope system
- Extracted time-frequency features for classification
- Targeted early detection of valvular heart disease
- **Tools:** ESP32, embedded system integration

## 3. Brain Tumor Segmentation Using Active Contour (2023)

- Implemented active contour (snake model) for biomedical image segmentation.
- Applied image processing techniques for boundary detection and region extraction.
- Improved segmentation accuracy through preprocessing and parameter tuning.
- **Tools:** MATLAB

## 4. Andromorphic Robotic Arm (EMG & Remote Control)

- Developed a dual-mode prosthetic robotic arm controlled using EMG signals and a web-based remote interface.
- Integrated ESP32-based embedded system for real-time control and actuation.
- Enabled both muscle-based gesture recognition and manual override functionality.
- **Tools:** ESP32, EMG sensors, Arduino ecosystem, web interface (HTML/CSS/JS)

## 5. Microfluidic Lab-on-Chip System

- Designed a microfluidic lab-on-chip platform to study fluid flow and drug transport across a membrane barrier.
- Fabricated the microfluidic device using PDMS through molding and casting techniques.
- Simulated blood/plasma flow and drug diffusion in controlled microchannels.
- Investigated material properties and cross-membrane interaction for biomedical applications.
- **Tools:** ANSYS

## 6. Identification of Potential Drug Targets in HTLV-1 via Integrated In Silico Approaches

- Conducted computational analysis to identify potential drug targets in HTLV-1 using integrated in silico methods.
- Performed bioinformatics-based screening and molecular docking studies to evaluate viral protein interactions.
- Analyzed drug–target binding affinity to support antiviral drug discovery research.
- **Tools:** AutoDock, PyRx, BLAST, Discovery Studio, GROMACS

# WEB DEVELOPMENT PROJECTS

## 1. Einthoven Club Official Website

- Developed and contributed to the official website of the Biomedical Engineering club at Military Institute of Science and Technology.
- **Link:** <https://einthovenclub.mist.ac.bd/>

## 2. AI-Powered Web-Based Rehabilitation Exercise Tracker (2025–2026)

- Built AI-based rehabilitation tracking system using pose and hand detection
- Used computer vision and MediaPipe for real-time posture correction
- Designed for remote physiotherapy applications
- **Link:** [https://sagorghj253.github.io/sagor\\_rehabilitation\\_project\\_BME410/](https://sagorghj253.github.io/sagor_rehabilitation_project_BME410/)

## 3. Andromorphic Robotic Arm – Remote Control Interface

- Designed a web interface for real-time remote control of a prosthetic robotic arm system.
- Integrated with ESP32-based hardware for dual-mode (EMG + web) control.
- Includes authentication-based access (username(“sagor”)/password(“46”) system).
- **Link:** <https://oishy17.github.io/hand/>

## 4. Digital Heart Murmur Detection System (2024)

- Developed signal processing-based stethoscope system
- Extracted time-frequency features for classification
- Targeted early detection of valvular heart disease

# PUBLICATIONS / CONFERENCE PAPERS

- Breast Cancer Subtype Prediction using KNN with PCA Feature Selection (2025) – Manuscript in preparation
- Identification of Potential Drug Targets in HTLV-1 via Integrated In Silico Approaches (2025) – Manuscript in preparation

# Technical Skills

## Programming Language

- C Programming
- Python (Basic)
- Linux Command Line
- MATLAB (Signal & Image Processing)
- Web Development: HTML, CSS, JavaScript

## Simulation & Design Tools

- SolidWorks (3D CAD design and modeling)
- Proteus (circuit design - Basic)
- COMSOL Multiphysics
- OpenSim (Biomechanical Modeling)
- Kimera (3D Scene Reconstruction)

## Data analysis

- SPSS

## Simulation & Computational Tools

- GROMACS (molecular dynamics simulation)
- ANSYS (fluid dynamics and engineering simulation)
- AutoDock /PyRx (molecular docking)
- PyMOL (molecular visualization)
- Discovery Studio (protein-ligand interaction analysis)
- Clustal Omega (sequence alignment)
- BLAST, UniProt, PDB
- Machine Learning (KNN, PCA)
- Computer Vision (Pose Detection)

## Hardware & Embedded Systems

- ESP32 microcontroller
- Arduino platform
- Biomedical sensors

## LABORATORY & TECHNICAL SKILLS

- Embedded systems development and biomedical device integration
- Rehabilitation engineering and assistive technology development
- Biochemistry laboratory techniques
- Biomaterials
- Biomechanics
- Molecular biology
- Biomedical instrumentation and biosignal acquisition
- Signal processing
- Biomedical image processing
- Wet lab and dry lab research methodologies

## EXPERIENCE

### Internship Trainee

- **National Institute of Biotechnology**  
Duration: May 12, 2025 – June 4, 2025

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

### CLUB CO-ORDINATOR

#### MIST Eindhoven Club

Year: 2026

- Organized a successful biomedical-focused event, “Celestra ’26”, promoting innovation and interdisciplinary collaboration in biomedical engineering.

## CERTIFICATIONS & TRAININGS

- **Champion, Poster Presentation Competition 2025**
  - Military Institute of Science and Technology presented the poster “Andromorphic Robotic Hand,” focusing on EMG and remote control systems
- Hands-on Training in Next-Generation Sequencing (NGS) – National Institute of Biotechnology (2025)
- Bioinformatics and Computational Drug Discovery Workshop – NIB (2025)
- MIST Eindhoven Club Presidential Panel Certification CELESTRA’26 (2026)

## LANGUAGES

- **Bengali** — Native Proficiency
- **English** — Professional Working Proficiency